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Transactive Memory Systems and Firm Performance: An Upper Echelons Perspective

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A substantial body of research uses the concept of transactive memory systems to describe, explain, and predict the behavior and performance of teams. In a multirespondent study of 99 small to midsized technology-based firms, we extend the concept into the unique context of top management teams and discuss its implications for firm performance. Building on the multifunctional and boundary-spanning role of top managers, we develop a novel theoretical account of how the performance implications of transactive memory are shaped by the individual and conjoint influences of a top management team's external social network ties and the rate of dynamism in the firm's competitive environment. In so doing, we link top management team transactive memory to firm performance through transformation—more than through application—of existing scholarly understanding and through distinct operating mechanisms informed by an upper echelons perspective of the firm. Our theory and supportive findings provide new evidence on the relationship between transactive memory and firm performance. We conclude by tracing the implications of our findings for upper echelons and transactive memory research.

Keywords: upper echelons theory; top management teams; transactive memory systems
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1. Introduction

Transactive memory is widely discussed as a useful concept for specifying and capturing the cognitive structure and functioning of teams (Ren and Argote 2011). Defined as a system for the shared division of cognitive labor with respect to encoding, storing, and retrieving knowledge from individual areas of expertise, transactive memory has been applied to understand, explain, and predict the behavior and performance of teams in a variety of settings (e.g., Austin 2003, Faraj and Sproull 2000, Littlepage et al. 2008, Zheng 2012). Transactive memory is argued to contribute to the effective functioning and performance of teams by enabling the efficient storage and retrieval of knowledge, optimal use of members' expertise, effective problem solving and decision making, efficient task coordination, and a reduced need for extensive communication (Lewis and Herndon 2011).

Despite the widespread application of transactive memory, its implications for understanding the behavior and performance of top management teams (TMTs)—a central unit of analysis in organizational research—has largely escaped scholars' attention. This is surprising not just because transactive memory provides a particularly useful lens for understanding the behavior and performance of teams engaged in complex and dynamic tasks but also because a central tenet of the upper echelons perspective is that the cognitive processes of TMTs have significant implications for organizational behavior and performance (Hambrick and Mason 1984). Most, if not all, core decisions, actions, and initiatives of TMTs are inextricably associated with externally driven imperatives, such as changes in market demand and shifts in technology. In their boundary-spanning role, which Organ (1971) referred to as the "linking pin" between the organization and its environment, top managers need to coordinate, expand, and use their cognitive resources to align the organization with the current and expected external environment (Daft and Weick 1984). Because the conceptualization and measurement of TMT cognitions are most often assumed or inferred from observable demographic characteristics such as functional background, education level, and tenure (Buyl et al. 2011, Cannella et al. 2008), the lack of direct studies and insights into TMT cognitions has contributed to a somewhat ambiguous understanding of how and when TMTs are actually shaping firm performance (Narayanan et al. 2011).

We develop a model that specifies the operational mechanisms of transactive memory and its impact on the multifaceted boundary-spanning conditions and constraints of TMTs. Our core contention is that the potentially beneficial influence of a TMT's transactive memory cannot be separated from the team's salient external knowledge sources and demands. Without relevant external knowledge flows, transactive memory is likely to grow increasingly insular and inward looking, which can hinder the ability of TMTs to maintain organizational alignment. Studies of executive scanning and decision making show that of all sources of external knowledge available top managers rely disproportionately on strong personal ties for accessing and absorbing private and preferential knowledge and insights (Aguilar 1967, Geletkanycz and Hambrick 1997, McDonald and Westphal 2003, McDonald et al. 2008). Most importantly, however, we argue that the contribution of a TMT's transactive memory to firm performance—alone and in combination with strong external networks—is likely to depend on the knowledge demands of the market and technological environment. Dynamic environments are difficult to analyze and frequently give rise to knowledge demands that exceed the cognitive abilities and wherewithal of individual top managers (Forbes 2007, Miller 2008). Thus, in dynamic settings, a TMT's transactive memory is all the more critical in mitigating cognitive overload and simplification by helping to harness the full range of members' cognitive resources.

We test these arguments using a multirespondent sample of small- to medium-sized enterprises operating in technology-based settings. In terms of organizational behavior and outcomes, smaller firms are likely impacted directly by top managerial cognition compared with larger organizations in which cognitive effects might be confounded by a variety of ecological and governance influences (Lubatkin et al. 2006). Moreover, the technology-based setting places a premium on cognitive demands and interdependency among top managers (Akgun et al. 2006, Ren and Argote 2011, Ren et al. 2006). We gather chief executive officer (CEO) and TMT data to directly measure each TMT's transactive memory, as well as both *perceptual* and *objective* measures of firm performance for robust testing. We find general support for our hypotheses but perhaps more importantly for the argument that the influence of a TMT's transactive memory on performance—in combination with the strength of external ties—is shaped by dynamism in the firm's environment.

Our study represents a rare extension of transactive memory research into new theoretical territory. In a novel departure from extant research, we extend theoretical understanding of the relationship between transactive memory and firm performance, developing new insights into the contingencies and constraints of transactive memory's effects within an upper echelons setting. Our findings shed new light on not only the conditions under which transactive memory is likely to be more or less effectual but also the situations in which transactive memory can be dysfunctional. In dynamic settings, for example, we find transactive memory is negatively associated with performance when top managers fail to maintain strong external ties with actors and institutions in the external environment. Without an influx of private and preferential knowledge, transactive memory may serve to ossify rather than promote adaptive organizational alignment in dynamic settings. For transactive memory researchers, these findings suggest that future inquiries need to situate transactive memory within the context of the team's social and task environment. For upper echelons researchers, we offer renewed theoretical insights into the influence of managerial cognition on firm performance, emphasizing a distributed rather than shared notion of cognition and enhancing the understanding of when, and under what conditions, TMT cognition is more (or less) likely to shape performance.

2. Transactive Memory Systems in Top Management Teams

The study of transactive memory is concerned with the explanation of team behavior and outcomes through an understanding of teams' information processing and structuring (Wegner 1987). Based on the insight that other people can be locations of knowledge and can be used as memory aides to supplement individual memory, transactive memory within teams emerges as team members come to rely on one another for complementary areas of knowledge (Lewis et al. 2005). As a system of cognitive interdependence, transactive memory is strengthened as team members learn about each other's areas of expertise and become "interdependent for encoding, storing, and retrieving information" (Peltokorpi 2008, p. 379). As a collective property of teams, transactive memory is manifested in a number of core underlying dimensions. In an experimental study on the effects of group training on performance, Liang et al. (1995) observed three behavioral manifestations of a well-developed transactive memory: memory differentiation (the tendency for members to specialize in different areas of knowledge), task credibility (the degree to which members' trust each other's knowledge), and task coordination (the ease of collaboration among team members in exchanging knowledge). Extending these findings from laboratory to field settings, Lewis (2003) discussed a team's transactive memory development as a reflective concept that gives rise to differentiated knowledge across team members (specialization), mutual trust in team members' knowledge (credibility), and harmonized knowledge processing within the team (coordination). Thus, specialization, credibility, and coordination represent manifest dimensions that one "would expect to observe in a group, if the group had developed TMS [transactive memory systems]" (Lewis and Herndon 2011, p. 1257). Importantly, transactive memory cannot be inferred from the presence of a single manifest dimension or by invoking each dimension in isolation. Rather, it is defined and measured as an emergent system of shared division of cognitive labor that concurrently gives rise to specialized knowledge among team members, mutual trust in the credibility of team members' knowledge, and coordinated processing of knowledge (Lewis 2003). By extension, a team's transactive memory is not traceable or reducible to what any one individual does (or does not), nor can it be found somewhere "between" individuals; rather, "it is a property of a group" (Wegner 1987, p. 191).

Whereas one strand of research has focused on the formation and antecedents of transactive memory, a second prominent one concerns its implications for team performance and outcomes. Prior studies demonstrate that transactive memory is linked to team performance and outcomes in a variety of settings (e.g., Austin 2003, Faraj and Sproull 2000, Heavey and Simsek 2014, Lewis 2004, Littlepage et al. 2008, Zheng 2012). Yet the implications of TMS development for TMTs, their work, and firm performance remain largely uncharted. We are aware of only a few empirical studies that have examined transactive memory among a population of executives: Rau's (2005) study of how sociopolitical relations (such as conflict and trust) among members of a TMT moderate the performance effects of certain dimensions of transactive memory (i.e., expertise composition and location awareness) and Rau's (2006) study of the associations among transactive memory dimensions (expertise composition and location), information gathering, and perceptual accuracy. Most recently, Heavey and Simsek (2014) found that the development of transactive memory in TMTs is positively associated with the firm's pursuit of an ambidextrous orientation. Although these studies have advanced our understanding of transactive memory within TMTs, further theoretical development of transactive memory in the unique context of TMTs is important for several reasons.

Specific attention to transactive memory in an upper echelons setting is warranted, not least for the practical reason that cognitive processes at this level result in the most salient consequences—positive or negative—for the entire organization but also because of the new theory and understanding it can bring to upper echelons and transactive memory research. In particular, TMTs can help stimulate novel thinking about the implications of transactive memory that may not be present for understanding its influence in other types of team settings. TMTs differ from other types of work teams (such as product and research and development (R&D) teams) in some core respects: (1) they are concerned with understanding and aligning both the external and internal environments of the organization; (2) they are generally embedded in ambiguity, complexity, and information overload; (3) their task is multifunctional; and (4) they rely on intermediaries for managing the daily affairs of the organization (Hambrick 1989). However, in managing these activities, TMTs are subject to an ongoing array of forces that gradually and subtly pull the team apart—what Hambrick (1994) termed "centrifugal forces." This occurs not only as a product of conflicts and turf wars but also because top managers often act as semiautonomous agents as a result of vested interest in unit/divisional performance (Hambrick 1995). It follows that TMTs represent a novel and unique context in which the influence of transactive memory is probably less certain and more complicated than in other types of teams.

Although the conceptual grounding and nature of the transactive memory construct is well established, and even as previous research has brought us closer to understanding some of its consequences for team sociopolitical processes, behaviors, and outcomes, its influence on firm behavior—and particularly performance—remains uncharted. Although Rau's (2005, 2006) work has provided important insights, because, "considered separately, the specialization, credibility, and coordination variables do not imply a TMS exists" (Lewis and Herndon 2011, p. 1257), we have yet to learn whether transactive memory as a reflective latent construct is associated with firm performance. By synthesizing insights concerning the performance effects of transactive memory from other team settings within an upper echelons context, we seek to understand how and when transactive memory can be consequential for firm performance. Put differently, to the extent that transactive memory can be consequential for the core work of TMTs, the firm's performance may also be impacted in ways not fully considered by researchers. Indeed, we argue that the positive impact of a TMT's transactive memory on firm performance is codetermined by the strength of the team's ties with external actors, together with the rate of dynamism in the environment. Next, we ground the study's specific hypotheses.

3. Hypotheses

3.1. The Impact of TMT's Transactive Memory on Firm Performance

Upper echelons theory posits that top managers shape firm outcomes through the processes by which they scan and interpret their environment and then act upon those interpretations. Thus, how TMTs organize and harness the insight and knowledge of their team members is central to their performance. A TMT's transactive memory expands the repertoire of cognitive resources available to the team because the shared responsibility for different kinds of knowledge across members allows each member to enlarge knowledge in a specific area while maintaining access to the knowledge of others (Hollingshead 2000, Wegner 1987). Moreover, because transactive memory allows for an attentional division of labor, TMTs with stronger transactive memory should display a more inclusive field of vision in environmental scanning and attend to a greater proportion of competitive cues in decision making and strategy formation than teams with weaker transactive memory. Just like air traffic controllers, TMTs with a well-developed transactive memory can allocate different portions of the competitive radar screen to managers based on knowledge and expertise. By reducing the duplication of cognitive effort and the cognitive burden of each individual manager, TMT transactive memory helps to better organize and harness attentional resources so as to reduce the risk of overlooking rivals, competitively relevant segments, or other significant competitive cues. These effects are beneficial because TMTs can exhibit myopic tendencies toward the known, familiar, and proximate, significantly inhibiting the firm's flexibility and responsiveness to emerging competitive developments (Ahuja and Lampert 2001, Shimizu and Hitt 2004). With a strong transactive memory, a TMT may reduce the risk of being overwhelmed by competitive stimuli and therefore succumbing to competitive blind spots and other types of performance-threatening "satisficing."

Aside from allowing a more encompassing field of vision and helping to avoid costly competitive blind spots, a well-developed transactive memory may also play an important role in expanding the interpretive repertoire of the TMT. By *interpretive repertoire*, we mean access to different interpretations and perspectives of the competitive environment. The distributive allocation of top managers' attention and cognitions to different portions of the environment facilitates specialized, in-depth, and nuanced interpretations of the market and technological environment, relative to a single shared conception. Consideration of multiple perspectives can help TMTs to see opportunities beyond what would be apparent to them individually (Heavey and Simsek 2014). This prevents the development of "groupthink," or decision making characterized by a strong emphasis on unanimity and redundant mental models (Janis 1982). By facilitating the "heedful" exchange and integration of top managers' individualized knowledge, expertise, and interpretations (Weick and Roberts 1993), transactive memory can facilitate the creation of higher-order knowledge that represents the "topic, theme, or gist" of lower-order information (Wegner et al. 1985, p. 264), thus permitting novel competitive interpretations that might not otherwise exist. By contrast, TMTs that do not make full use of members' knowledge and interpretations are less likely to integrate unique expertise and fully exploit different interpretive schemes to creatively respond to competitive situations. Even as members of these teams may hold complementary pockets of knowledge, transactive memory provides the "connective tissue" that links top managers' otherwise isolated and private pockets of knowledge.

Taken together, these arguments suggest that by facilitating a cooperative and extensive division of cognitive labor, together with routines for the exchange and integration of knowledge, a TMT's transactive memory can enable the team to handle complex demands and interdependency more efficiently than it would otherwise. TMTs with high levels of transactive memory are likely to utilize each member's expertise and resources, as well as deepen individual knowledge. This gives them an advantage over teams with lower levels of transactive memory in bringing a greater range of relevant knowledge and interpretations to boundary-spanning tasks such as environmental scanning and search. In this way, transactive memory can promote access to diverse cognitive resources essential for making effective strategic decisions, and it provides the requisite level of cognitive content and coordination for TMTs to perform complex tasks and meet ambiguous job demands. Thus, we predict the following.

Hypothesis 1. *There is a positive relationship between the strength of a top management team's transactive memory and firm performance.*

3.2. The Contingent Influence of TMT External Ties

Even though a well-developed transactive memory can provide a TMT with superior knowledge processes, few TMTs hold all the insights and interpretations required to cope with the demands of the environment. Consequently, they often rely on network ties to fulfill their knowledge gaps. Although we expect that firm performance is shaped by a TMT's transactive memory, we elaborate this insight here by suggesting that the degree of this shaping influence depends on the richness, relevance, and timeliness of external knowledge flows into the team from ties with actors outside the firm. Defined as the average duration, frequency, and emotional intensity of external ties (Granovetter 1973, Smith et al. 2005), the strength of TMT's ties with external actors plays a large part in shaping their access to private and preferential market knowledge (Ingram and Roberts 2000), as well as their ability to absorb complex, tacit, and nonstandardized knowledge (Hansen 1999). Compared with publicly available information, private information is "soft" and "references idiosyncratic and non-standardized information" (Uzzi and Lancaster 2003, p. 384), thereby providing a more advantageous basis for managerial decision making and action. Moreover, even if a TMT has preferential access to external knowledge, it must have the ability to combine existing team knowledge with external knowledge for greater effectiveness. Strong external ties maintained by top managers enhance the knowledge-absorption effects afforded by differential specialization of transactive memory. We say this in part because managers engaging in differential specialization can engage in more intensive boundary spanning when they have strong ties outside the firm. Managers can utilize these ties for a variety of functions beyond merely scanning the environment and keeping informed of developments. They may also find them useful in probing elements of the competitive environment, analyzing and interpreting events, and seeking feedback on initiatives (Ancona and Caldwell 1992). Under such conditions, the performance manifestations of transactive processes are more salient not only because specialized and shared managerial understandings are likely more grounded in novel insights and interpretations of the environment but also because transactive memory serves to integrate these understandings.

Without strong external ties, the likely contribution of a TMT's transactive memory system to firm performance is diminished. Although a lack of ties altogether insulates the team from relevant external knowledge and results in an emphasis on internal and potentially insular knowledge, a failure to build and maintain strong ties may also be detrimental. Without strong ties, a TMT is less privy to private, preferential, and fine-grained knowledge from actors in the value chain and stakeholder network. Customers, for example, are less likely to take the time to explain their needs and problems, suppliers less willing to impart fine-grained knowledge of new technologies, and allies and competitors are ill-inclined to share mutually beneficial value-creating business models. Even when available through other means, the ability of the team to fully probe, make sense of, and assimilate information is limited by the absence of strong ties that could confer nuanced meanings and understandings. Consequently, the value of knowledge exchanged and developed in a TMS provides a less rich and distinctive source of advantage compared with TMTs that have formed strong external ties. Thus, we predict the following.

Hypothesis 2. *The positive association between a top management team's transactive memory and firm performance will be stronger as the average strength of the top management team's external ties increases.*

3.3. The Contingent Influence of Environmental Dynamism

Although many factors shape the knowledge demands of the environment (Huber and Daft 1987), the level of dynamism—defined as the perceived speed and unpredictability of demand and technological change—imposes considerable cognitive challenges on top managers (Anderson and Tushman 2001, Atuahene-Gima and Li 2004). Dynamic environments are cognitively hostile in that they require more strategic decision making per unit of time than top managers can ordinarily process (Miller 2008). Owing to the rapid and unpredictable changes in demand and/or technologies, dynamic environments rapidly render current managerial knowledge, experience, and learning obsolete (Anderson and Tushman 2001, Miller 2008). For example, Anderson and Tushman (2001) found that industry dynamism, rather than complexity or munificence, was associated with higher rates of mortality among firms because dynamic settings often necessitate the "unlearning" of past lessons. Similarly, Henderson et al. (2006) found that peak performance occurred earlier for CEOs in dynamic industries than in stable ones, since "environmental upheaval is often so great as to make yesterday's knowledge useless" (p. 451). At minimum, top managers operating in such settings must update their knowledge of technological and market conditions on an ongoing basis.

Consequently, we expect the performance effects of a TMT's transactive memory to be heightened in dynamic environments since transactive memory enables the team to organize and enlarge its information-processing capacity to cope with the knowledge demands of the environment. Because managers specializing in different domains of knowledge will likely view the environment differently, a TMT's transactive memory helps to preserve the requisite variety needed to make sense of and address uncertainty in the environment (Kilduff et al. 2000). The performance implications of a TMT's transactive memory may also be amplified in dynamic settings since routines for differential specialization and coordination enable each manager to update knowledge and subsequently transact that updated knowledge with each other. In dynamic settings, transactive memory can thus play an adaptive function by providing a cognitive structure for knowledge renewal without overly taxing each individual manager. Conversely, the performance effects of a TMT's transactive memory are likely to be less evident in stable settings, where there is less need for top managers to rapidly renew and update their knowledge stock.

Finally, the performance and growth benefits of a TMT's transactive memory may also be more manifest in dynamic than in stable settings because the focused and efficient nature of team interaction can give the team a speed advantage, which is a key competitive attribute when markets and technologies are in a state of flux. Compared with stable settings where the direction of market and technological change is relatively certain and predictable, dynamic settings represent ill-structured "moving targets" for TMTs. Thus, because dynamic environments place a premium on the timely exploitation of fleeting opportunities (Williams 1992) and call for speed in strategic decision making and execution (Eisenhardt 1989), a TMT's transactive memory should exhibit a stronger association with performance by contributing to the speed and deftness with which top managers assemble and transact knowledge. By contrast, the advantages of a TMT's transactive memory in facilitating timely collective action are less likely to benefit performance in stable settings that do not require the same degree of speed and timely action, and predefined mechanisms can be designed and deployed for various contingencies (Faraj and Xiao 2006). Thus, we deduct the following.

Hypothesis 3. *The positive association between a top management team's transactive memory and firm performance will be stronger as environmental dynamism increases.*

3.4. Cocontingency of TMT External Ties and Environmental Dynamism

Although we hypothesized that the strength of managerial networks outside the firm serves to reinforce and strengthen the beneficial influence of a TMT's transactive memory by infusing the team with rich, relevant, and timely knowledge, we expect that the contingent importance of external ties itself depends on extent of knowledge demands vis-à-vis the level of dynamism in the environment. The enrichment effect of strong ties in infusing transactive memory with updated interpretations and insights is less likely to manifest itself in stable settings. When demand and technology are perceived as stable, top managers have little to gain from reaching outside the team for renewed insights and interpretations because their stock of existing knowledge and its periodic updating, embedded in their transactive memory, may be sufficient to match the knowledge demands of the environment. Indeed, under such circumstances, we suspect managers are less likely to intensively utilize their networks to update their knowledge.

Because dynamic environments are characterized by the creative destruction of demand and technological conditions, the need for managers to unlearn the lessons of the past and update their knowledge and frames of reference is amplified (Anderson and Tushman 2001). Under these circumstances, the knowledge transacted inside the team may be insufficient to match the informational demands of the environment. Since managers rely heavily on their networks to make sense of uncertainty and update their understanding of the environment, the performance implications of transactive memory are likely more closely tied to the richness of knowledge flows from the external environment vis-à-vis the strength of external ties. We say this because in dynamic environments top managers need to maintain constant "traffic" with their environment, "importing and exporting ideas and materials, managing impressions, and scanning the environment for potential opportunities and threats" (Mehra et al. 2006, p. 65). Thus, strong external ties can enhance the operation of a TMT's transactive memory by enabling each manager to update their knowledge while imparting renewed frames of reference for interpreting the environment. Although weak ties provide different insights into *what* is happening in dynamic environments, the guarded nature of such exchanges often precludes distinctive and deep insights into why it is happening or what might happen next. As such, without a strong system of external ties, a TMT's transactive memory may become disconnected from external conditions, which might even hinder performance by serving to transact outmoded knowledge and interpretations.

These arguments suggest that the influence of a TMT's transactive memory and the complementary role of strong ties to external actors cannot be separated from the level of environmental dynamism in which the TMT operates. As the level of environmental dynamism increases, transactive memory needs to be coupled with externally strong ties to ensure the ongoing accuracy and relevance of knowledge content. However, at low levels of dynamism, such coupling is of lesser importance for firm performance. Thus, our final prediction is as follows.

Hypothesis 4. *At a high level of environmental dynamism, the joint effects of a top management team's transactive memory and average strength of external ties on firm performance will be stronger than at a lower level of environmental dynamism.*

4. Method

4.1. Sampling and Data Collection

Using CorpTech (now part of InfoUSA), we built a sampling frame of 467 small- to medium-sized enterprises (SMEs) operating in technology-based sectors in the northeastern United States. We chose small- to medium-sized enterprises because firms of this size (i.e., employing 50 to 500 individuals) and age (i.e., considerably older than start-ups) represent a direct litmus test of our hypotheses. Performance in larger firms, by contrast, can often be driven by a diverse set of influences extraneous to TMTs, such as multiple product lines, complex management systems, and governance mechanisms (Lubatkin et al. 2006). Since SMEs have fewer hierarchical levels, their top managers are more likely to serve both operational and strategic roles. As a result, the setting for testing the influence of transactive memory on firm performance is generally less confounded. We chose a sample of technology-based firms, defined as those that "emphasize invention and innovation in business strategy, deploy a significant percentage of their financial resources to R&D, employ a relatively high percentage of scientists and engineers, and compete in worldwide, short-life-cycle product markets" (Milkovich 1987, p. 104). The firms in the sample operated in a diverse mix of technologically focused industries such as advanced materials, hardware and components, software, and telecommunications and Internet.

We initially sent a letter to each of the CEOs in our sample inviting them to participate in the study.

Within two weeks of sending the letter, we telephoned the CEOs to secure participation and schedule a meeting to administer the survey. One advantage of administering the survey on-site is that it allowed the authors to build enough rapport with the CEOs to encourage them to distribute the survey to their team. All told, for 99 firms, we received responses from the CEO and at least another member of the top management team. The average intrateam response rate was 57.4% across the sample. In the majority of cases (70/99), we received completed surveys from the CEO and at least two additional executives. We found no systematic differences between responding and nonresponding firms in terms of age, size, sales, or employment growth using both t-tests and two-sample Kolmogorov–Smirnov tests. To decipher any systematic differences with respect to transactive memory, we used t-tests and a logistic regression function to compare fully responding (responses from CEO and TMT) and partially responding (response from CEO only) firms. We found no evidence of a systematic bias or difference.

4.2. Measurement

Apart from the dependent variable, which relies on the assessment of the CEO, our measures involved the aggregated responses of top managers. We separated the measurement of our independent and dependent variables using separate respondents to minimize the influence of common method effects. To warrant aggregation, we computed indices of interrater agreement and interrater reliability: the r_{wg} index to measure interrater agreement (James et al. 1984, 1993) and intraclass correlations (ICCs) to measure interrater reliability (LeBreton and Senter 2008).

4.2.1. Independent Variable. We measured transactive memory using Lewis's (2003) 15-item scale. This measure is particularly recommended when predicting the effects or outcomes of transactive memory in organizational settings (Heavey and Simsek 2014, Lewis and Herndon 2011). Following previous studies, we used only TMT member (excluding CEO) ratings of transactive memory (although our results remain unchanged when the CEO is included) to guard against common method bias. To test the structure of transactive memory as a higher-order factor with subdimensions of specialization, credibility, and coordination, we conducted a confirmatory factor analysis (CFA). Consistent with earlier studies (Heavey and Simsek 2014, Lewis 2003), the results of the analysis suggested acceptable fit ($\chi^2(80) = 201$; comparative fit index (CFI) $= 0.91$; Tucker–Lewis index (TLI) $= 0.88$; incremental fit index (IFI) $= 0.91$; root mean square error of approximation, 0.09). The coefficient α for the entire 15-item scale was 0.86. An acceptable level of interrater agreement (median $r_{wg(j)} = 0.94$) and reliability (ICC1 $= 0.31$, ICC2 $= 0.69$) justified the aggregation of top managers' responses to the team level of analysis (Chan 1998).

4.2.2. Moderating Variables. To measure the average strength of the TMT's external ties, we followed the methodology developed by Collins and Clark (2003) and commonly employed by upper echelons researchers (Cao et al. 2010, Heavey et al. 2015, Smith et al. 2005). This involves assessments of the three dimensions of tie strength: frequency, duration, and emotional intensity (Granovetter 1973). Respondents were first asked to indicate the number of individual contacts they had in each of nine external categories: board members, suppliers, customers, financial institutions, competitors, alliance partners, governmental agencies, trade associations, and other. To measure frequency of communication, respondents were asked to indicate, on average for each category of contact, how many they interacted with once a month. To measure duration, respondents were asked to evaluate how long, on average, they had known these contacts. Finally, to measure emotional intensity, respondents were asked to indicate, on average for each category of contact, the closeness of their relationship. A five-point Likert scale was used to gauge emotional intensity (with 1 indicating "not at all close" and 5 indicating "extremely close"). Following Collins and Clark, the strength of external ties was measured as the linear combination of the standardized scores of the three components of tie strength. As a partial verification of this measure, we found that the average strength of external ties was positively associated with TMT age ($r = 0.35$, $p < 0.01$), firm tenure ($r = 0.36$, $p < 0.01$), and industry tenure ($r = 0.36$, $p < 0.01$). Because our items referenced the network ties of individual executives, using measures of agreement such as r_{wg} to justify aggregation is not appropriate because networks can be expected to vary across individuals, even within the same team. Aggregation is justified on two other bases instead. First, because all individuals are identified by the CEO as belonging to the TMT, their responses can reasonably be combined to gain a measure of the average strength of the TMT's external ties (Heavey et al. 2015). Second, to gain some insight into the reliability of the average strength of ties, we relied on intraclass correlation scores (ICC1 and ICC2), which are computed to provide estimates of the reliability of the group mean (Bliese 2000). The ICC1 and ICC2 for the average strength of external ties across team members are 0.11 and 0.58, respectively, providing "prima facie evidence a group effect" for the average strength of external ties (LeBreton and Senter 2008, p. 838).

We measured environmental dynamism using a seven-item Likert scale, developed and validated by Atuahene-Gima and Li (2004), which has been used in a number of prior studies (e.g., Danneels and Sethi 2011). The first four items capture the extent of technological dynamism. Respondents were asked to evaluate the extent to which they agreed with the following assertions: (a) "the technology in our industry is changing quite rapidly,"

(b) "technological changes provide big opportunities for our industry," (c) "a large number of product ideas have been made possible through technological breakthroughs in our industry," and (d) "there have been major technological breakthroughs in our industry." The scale also included three items reflecting the degree of market dynamism, which asked respondents to rate the extent to which (a) "customer demand and product preferences change quite rapidly," (b) "customers tend to look for new products all the time," and (c) "we witnessed demand for our products from customers who never bought from us before." The scale exhibited satisfactory levels of interitem reliability ($\alpha = 0.88$), interrater agreement (median $r_{wg} = 0.95$), interrater reliability (ICC1 = 0.42, ICC2 = 0.70; $F = 3.343$, $p < 0.001$), and measurement properties ($\chi^2(13) = 89.004$, $p < 0.001$; CFI = 0.94, TLI = 0.91, IFI = 0.94).

4.2.3. Dependent Variable. Previous research suggests that capturing firm performance requires the use of multiple measures (Richard et al. 2009, Stam and Elfring 2008). Subjective, perceptual managerial assessments of performance are particularly useful for capturing broader dimensions of performance relative to direct competitors and have been shown to exhibit strong reliability and validity (Dess and Robinson 1984, Robinson and Pearce 1988). Objective measures of firm performance are less prone to common method bias, but they can be hard to obtain because privately held, smaller firms, such as those in our sample, are not legally required to publish accounting data. Given the unique strengths and weaknesses of each measure, we used both to test our hypotheses.

First, we used an adapted version of Stam and Elfring's (2008) multidimensional measure of firm performance consisting of eight items: sales growth, employment growth, market share, profit, innovation in products and services, quality of products and services, cost control, and customer satisfaction. Because CEOs in smaller firms are generally most knowledgeable about firm performance relative to competitors, CEOs were asked during the first author's interviews to rate their firm's performance (over the past year) on each item relative to competitors (much worse, about same, or much better) along a seven-point scale. The scale demonstrated acceptable internal consistency ($\alpha = 0.78$) and psychometric properties ($\chi^2(17) = 37$, $p < 0.01$; CFI = 0.92, IFI = 0.92, TLI = 0.87). Although perceptual performance measures have a long history in organizational studies (e.g., Robinson and Pearce 1988), and our particular measure has been validated previously by Stam and Elfring, we took three cautionary steps. First, we correlated this measure with an objective measure of sales provided by CorpTech. Our subjective measure was positively and significantly correlated with objective sales ($r = 0.25$, $p < 0.05$), providing some evidence of convergent validity. Second, we compared CEO and TMT evaluations of firm performance and found both evaluations were positively and significantly correlated ($r = 0.81$, $p < 0.001$) and exhibited high levels of interrater agreement and reliability (median $r_{wg(j)} = 0.97$; ICC1 = 0.59, ICC2 = 0.76; $F = 4.21$, $p < 0.001$). Finally, to guard against the possibility of any response bias, we compared the distribution of CEO and TMT performance evaluations using the two-sample Kolmogorov–Smirnov test. No differences were found.

Additionally, as an objective index of firm performance, we obtained from the CEO the rate of the firm's sales growth over the past year. As with other studies (e.g., He and Wong 2004), we focused on sales growth for several reasons. First, sales growth is among the most widely used measures of firm performance in research, particularly for assessing the success of new ventures and SMEs (e.g., Robinson and McDougall 2001). Sales growth provides a measure of how well an organization relates to its environment by successfully expanding product market scope (Dess and Robinson 1984). Second, unlike estimates of profitability, sales growth does not suffer from accounting measurement problems (He and Wong 2004). Finally, setting aside the fact that there exists no database reporting the performance of privately held firms,[1] sales growth tends to be a more reliable dimension of performance among these firms than other income-based measures because privately held firms may have incentives to minimize reported taxable income. During our interviews, we asked the CEOs to provide a precise estimate of sales growth during the past year. For this task, most CEOs consulted with financial documents and used a calculator to compute the sales growth. Several CEOs told the story behind sales growth, lending face validity to the measure. For example, the CEO of a company that designed, manufactured, and distributed springs explained how his firm increased sales by finding new markets for the core technology. We adjusted our sales growth measure for outliers in two alternative ways: by deleting the extreme outliers and by winsorizing extreme values to the outer tail of the distribution (Tabachnick and Fidell 1996). Although both approaches yielded almost identical results in terms of support for our hypotheses and patterns of interactions, we opted for the latter approach given our sampling constraints. The median sales growth in the sample was 2.5%. Of the 99 firms in our sample, approximately one-third experienced a decrease in sales growth, reflecting the macroeconomic context at the time of our study. Reporting of negative sales growth figures in a variety of cases mitigates social desirability concerns.

4.2.4. Statistical Controls. We controlled for variables at the firm and team levels of analysis. At the firm level, we controlled for firm age, size, and past performance. We controlled for firm size (i.e., the number of full-time employees), because even among SMEs, larger firms have more resource advantages than small firms that might affect their performance (Collins and Clark 2003). We also controlled for firm age (i.e., the number of years since the company was established), because it is associated with institutional routines and norms that might engender inertial behavior (Tushman and Romanelli 1985). Because current performance is partly a reflection of past performance, we controlled for past sales growth by asking the CEO to estimate the growth rate in company's sales during the past three years. As with our other sales growth measure, these data were obtained during our on-site personal interview with the CEO. In many cases, the CEO relied on both company documentation and a calculator to provide an accurate estimate of past sales growth.

Because our focus is on explaining the relationship between TMT transactive memory and firm performance, we controlled for several aspects of each team's composition. We first controlled for team size, since it provides for a parsimonious representation of the team's compositional and structural context (Amason and Sapienza 1997). We also controlled for the average age, tenure, and educational attainment of the team, since these compositional factors have been argued to shape the attentional and information-processing capacities of teams (Hambrick and Fukutomi 1991). Finally, to capture the heterogeneity of cognitive resources embedded within the team, and because a TMT's transactive memory might be driven by the functional expertise of executives, we controlled for the average level of intrapersonal functional diversity. To measure a TMT's intrafunctional diversity, each manager was asked to indicate the number of years they has worked in the following functional departments: sales, manufacturing, accounting/finance, personnel/human resources, distribution/logistics, R&D, legal, administrative, and general management. Following the procedures recommended by Bunderson and Sutcliffe (2002), we first estimated an intrapersonal functional diversity score for each top manager and then computed the average of these.

5. Analysis and Findings

5.1. Descriptive Statistics

We present the means, standard deviations, and Pearson correlations of all variables in Table 1. The highest correlation ($r = 0.62$) and variance inflation factor (VIF = 2.231) are below the recommended cutoffs of 0.65 for correlation (Tabachnick and Fidell 1996) and 10 for VIF (Neter et al. 1996). We performed two nested CFAs: one in which a TMT's transactive memory and performance were combined and one in which they were separated. We found that the two-factor model fit significantly better than the one-factor model, suggesting no problems of discriminant validity (Bagozzi et al. 1991, Kline 1998). To increase our confidence in this finding, we performed two additional tests. First, we performed Anderson and Gerbing's (1988) paired construct test with a constrained model, in which the covariance is constrained to 1.0, and an unconstrained model in which the covariance is freely estimated. We found that the unconstrained model significantly outperformed the constrained model ($\Delta\chi^2(1) = 57.7$, $p < 0.001$). Thus, the discriminant validity of transactive memory and firm performance constructs is supported. Second, we observed that the average variance explained by both transactive memory and performance exceeds the squared correlation between the two, thereby providing further evidence of discriminant validity (Fornell and Larcker 1981).

5.2. Hypothesis Testing

We tested our hypotheses using hierarchical moderated regression (see Table 2). As reported earlier, we

Table 1 Descriptive Statistics and Correlations

Variable	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12
1. *Sales growth* (past year)	0	1	1											
2. *Subjective performance* (past year)	0	1	0.35**	1										
3. *TMT transactive memory*	5.46	0.61	0.10	0.28*	1									
4. *TMT average strength of external ties*	0	1	0.00	0.03	0.03	1								
5. *Environmental dynamism*	4.37	0.95	0.25*	0.38**	−0.04	0.24*	1							
6. *Firm age*	49.5	34.9	−0.33**	−0.03	0.11	−0.04	−0.18†	1						
7. *Firm size*	157	217	−0.16	0.03	0.06	−0.04	0.20*	0.19†	1					
8. *Past performance* (past three years)	29.4	110	0.34**	0.14	−0.24*	−0.07	0.23*	−0.24*	−0.09	1				
9. *TMT size*	6.20	2.30	−0.14	−0.08	−0.01	0.04	−0.02	−0.08	−0.07	−0.05	1			
10. *TMT tenure*	15.5	7.66	−0.19	0.04	0.23*	0.36**	−0.02	0.32**	0.05	−0.18	−0.06	1		
11. *TMT education*	13.0	9.62	−0.07	−0.01	−0.01	−0.18†	−0.07	0.23*	0.10	0.00	−0.15	−0.01	1	
12. *TMT age*	51.8	6.3	−0.01	0.19†	0.24*	0.35**	0.09	0.20*	0.02	0.22*	−0.07	0.62**	0.03	1
13. *TMT functional diversity*	0.31	0.19	−0.04	−0.09	−0.14	0.05	−0.11	−0.02	−0.08	0.04	−0.05	0.03	−0.04	0.01

Note. $N = 99$.
†$p < 0.10$; *$p < 0.05$; **$p < 0.01$.

Table 2 Results of Hierarchical Regression Analysis for Performance Rating

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Control variables							
Firm age	−0.03 (1.26)	−0.05 (1.27)	0.02 (1.32)	0.05 (1.34)	0.05 (1.35)	0.04 (1.35)	0.05 (1.35)
Firm size	0.03 (1.08)	0.03 (1.08)	−0.07 (1.17)	−0.07 (1.17)	−0.07 (1.17)	−0.08 (1.19)	−0.06 (1.21)
Past performance	0.17 (1.10)	0.23* (1.14)	0.14 (1.22)	0.11 (1.24)	0.04 (1.75)	0.01 (1.80)	−0.05 (1.91)
TMT size	−0.03 (1.02)	−0.03 (1.02)	−0.04 (1.03)	−0.07 (1.04)	−0.08 (1.06)	−0.06 (1.08)	−0.01 (1.15)
TMT tenure	−0.14 (1.81)	−0.18 (1.84)	−0.17 (1.95)	−0.22† (2.00)	−0.23† (2.03)	−0.24† (2.03)	−0.22† (2.04)
TMT education	−0.05 (1.04)	−0.06 (1.04)	−0.05 (1.07)	−0.06 (1.08)	−0.06 (1.08)	−0.05 (1.10)	−0.02 (1.11)
TMT age	0.30* (1.70)	0.27* (1.71)	0.22† (1.78)	0.27* (1.84)	0.29* (1.87)	0.29* (1.87)	0.29* (1.89)
TMT functional diversity	−0.10 (1.02)	−0.05 (1.05)	−0.03 (1.05)	−0.04 (1.06)	−0.05 (1.07)	−0.06 (1.08)	−0.05 (1.09)
Main effects							
TMT transactive memory		0.31** (1.15)	0.32** (1.16)	0.33** (1.16)	0.33** (1.17)	0.32** (1.18)	0.17 (1.76)
TMT average strength of external ties			−0.08 (1.36)	−0.11 (1.34)	−0.13 (1.42)	−0.13 (1.42)	−0.11 (1.43)
Environmental dynamism			0.37** (1.30)	0.40*** (1.32)	0.42*** (1.36)	0.41*** (1.36)	0.41*** (1.36)
Two-way interaction effects							
TMT transactive memory × TMT average strength of external ties				0.25* (1.12)	0.32* (1.74)	0.31* (1.74)	0.23† (1.91)
TMT transactive memory × Environmental dynamism					−0.13 (1.88)	−0.13 (1.88)	−0.10 (1.92)
TMT average strength of external ties × Environmental dynamism						−0.13 (1.12)	−0.15 (1.12)
Three-way interaction effect							
TMT transactive memory × TMT average strength of external ties × Environmental dynamism							0.26* (1.79)
R^2	0.09	0.17	0.27	0.33	0.33	0.35	0.39
ΔR^2		0.08	0.10	0.06	0.00	0.02	0.04
ΔF		7.954**	5.661**	6.270**	1.013	1.802	4.446*

Notes. Values in parentheses are VIFs. $N = 99$.
$^{\dagger}p < 0.10$; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

relied on two different performance measures to test our hypotheses. For purposes of interpretability, we report the results of these two analyses in separate tables. Model 1 includes all the control variables, Model 2 includes the main effect of transactive memory, Model 3 includes the main effect of our two moderators (average strength of external ties and environmental dynamism), Models 4–6 include each of the three possible two-way interaction terms, and Model 7 includes the three-way interaction term. To detect possible multicollinearity, we calculated the VIF for all variables used in the regression analysis. The VIF values are reported below each of the regression coefficients in Tables 2 and 3. As shown in Tables 2 and 3 (Model 2), TMT transactive memory is positively associated with subjective performance ($\beta = 0.31$, $p < 0.01$) and sales growth ($\beta = 0.21$, $p < 0.05$), providing strong support for Hypothesis 1. Tables 2 and 3 (Model 4) show that Hypothesis 2 is supported for both subjective performance ($\beta = 0.25$, $p < 0.05$) and sales growth ($\beta = 0.21$, $p < 0.05$). We followed the procedures set out by Aiken and West (1991) and Cohen et al. (2003) to plot the relationship between TMT's transactive memory and both measures of performance based on conditional values of managerial external tie strength (i.e., ±1 standard deviation).[2] We standardized the independent and moderating variables prior to creating interaction terms (Dawson and Richter 2006). As shown in Figures 1 and 2, the effects of transactive memory on subjective performance and sales growth are more pronounced when TMTs maintain strong ties with external actors and institutions, such as customers, suppliers, and government. However, we did not find

Table 3 Results of Hierarchical Regression Analysis for Sales Growth

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Control variables							
Firm age	−0.23*	−0.24*	−0.19†	−0.17	−0.16	−0.17	−0.16
	(1.26)	(1.26)	(1.33)	(1.35)	(1.35)	(1.35)	(1.35)
Firm size	−0.10	−0.10	−0.16	−0.16	−0.16	−0.17	−0.15
	(1.07)	(1.07)	(1.17)	(1.17)	(1.17)	(1.20)	(1.21)
Past performance	0.27**	0.31**	0.26*	0.23*	0.25*	0.23†	0.16
	(1.10)	(1.14)	(1.22)	(1.24)	(1.84)	(1.90)	(2.02)
TMT size	−0.13	−0.13	−0.15	−0.16†	−0.16†	−0.15	−0.10
	(1.02)	(1.02)	(1.04)	(1.05)	(1.09)	(1.10)	(1.17)
TMT tenure	−0.28*	−0.31*	−0.31*	−0.35**	−0.35**	−0.35**	−0.33*
	(1.81)	(1.83)	(1.95)	(2.00)	(2.03)	(2.03)	(2.04)
TMT education	−0.09	−0.09	−0.07	−0.09	−0.09	−0.08	−0.05
	(1.04)	(1.04)	(1.07)	(1.08)	(1.08)	(1.10)	(1.12)
TMT age	0.27*	0.25*	0.22†	0.26*	0.26*	0.26*	0.26*
	(1.70)	(1.71)	(1.79)	(1.85)	(1.87)	(1.87)	(1.87)
TMT functional diversity	−0.08	−0.05	−0.04	−0.05	−0.05	−0.06	−0.04
	(1.02)	(1.05)	(1.05)	(1.06)	(1.07)	(1.08)	(1.09)
Main effects							
TMT transactive memory		0.21*	0.22*	0.23*	0.23*	0.22*	0.07
		(1.15)	(1.16)	(1.16)	(1.16)	(1.18)	(1.74)
TMT average strength of external ties			−0.02	−0.05	−0.04	−0.04	−0.02
			(1.36)	(1.38)	(1.42)	(1.42)	(1.43)
Environmental dynamism			0.20†	0.23*	0.22*	0.22*	0.21*
			(1.33)	(1.35)	(1.42)	(1.42)	(1.42)
Two-way interaction effects							
TMT transactive memory × TMT average strength of external ties				0.21*	0.18	0.18	0.09
				(1.12)	(1.78)	(1.78)	(1.95)
TMT transactive memory × Environmental dynamism					0.04	0.04	0.07
					(1.98)	(1.99)	(2.02)
TMT average strength of external ties × Environmental dynamism						−0.09	−0.11
						(1.12)	(1.12)
Three-way interaction effect							
TMT transactive memory × TMT average strength of external ties × Environmental dynamism							0.27*
							(1.78)
R^2	0.28	0.32	0.35	0.39	0.39	0.40	0.44
ΔR^2		0.04	0.03	0.04	0.00	0.01	0.04
ΔF		4.338*	1.867	4.706*	0.108	0.931	5.140*

Notes. Values in parentheses are VIFs. $N = 99$.
†$p < 0.10$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

support for our third hypothesis, which posited that environmental dynamism would serve to moderate the relationship between a TMT's transactive memory and firm performance. As shown in Tables 2 and 3 (Model 5), the interaction between a TMT's transactive memory and environmental dynamism is not statistically significant for either measure of performance.

Finally, we tested Hypothesis 4 by following the procedures of Aiken and West (1991) and Dawson and Richter (2006) for testing three-way interactions. Our hypothesis that the moderating effect of strength of external ties would vary across levels of dynamism was supported. As shown in Tables 2 and 3 (Model 7), the three-way interaction term is statistically significant for measures of both subjective performance ($\beta = 0.26$, $p < 0.05$) and sales growth ($\beta = 0.27$, $p < 0.05$). To gain further insight into the exact pattern of effects, we plotted the three-way interaction following the procedures outlined by Dawson and Richter (2006). As shown in Figures 3 and 4, the moderating effect of strength of external ties on the relationship between a TMT's transactive memory and firm performance varies across levels of dynamism. At low levels of dynamism (panel (a) in Figures 3 and 4), increases in the strength of external ties do not strengthen the magnitude of the relationship between transactive memory and performance, as evidenced by the parallel slopes at low and high levels of strength. However, at high levels of environmental dynamism (panel (b) in Figures 3 and 4), the influence of external ties becomes more prominent. As shown by the

Figure 1 **The Moderating Role of TMT Strength of External Ties on the Relationship Between TMT Transactive Memory and Subjectively Measured Firm Performance**

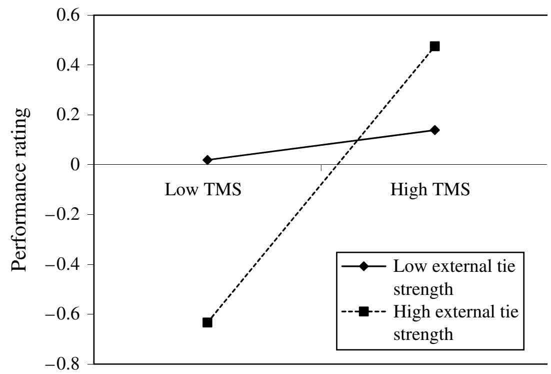
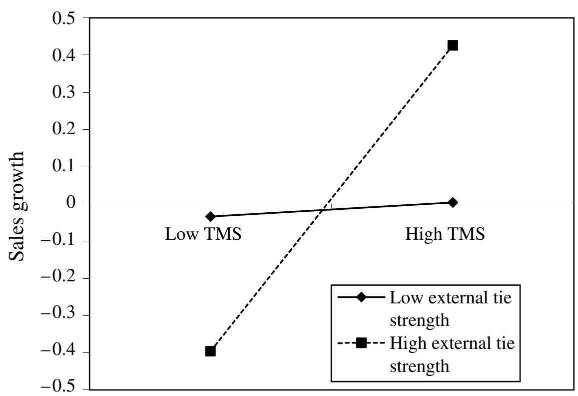


Figure 2 **The Moderating Role of TMT Strength of External Ties on the Relationship Between TMT Transactive Memory and Sales Growth**

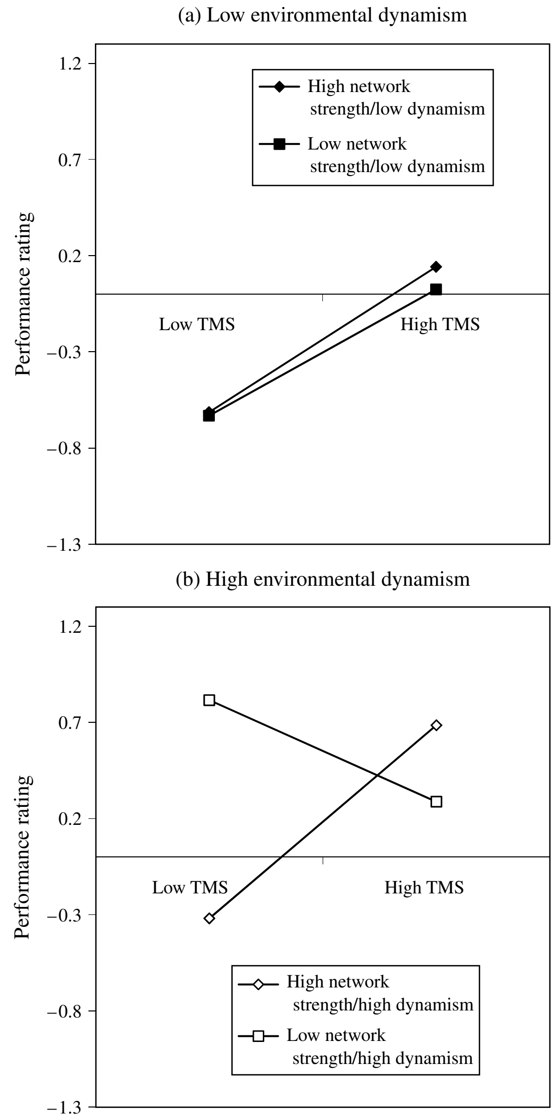


disordinal interaction, the association between a TMT's transactive memory and firm performance changes from negative to positive as the strength of external ties increases.

5.3. Tests for Common Method Bias

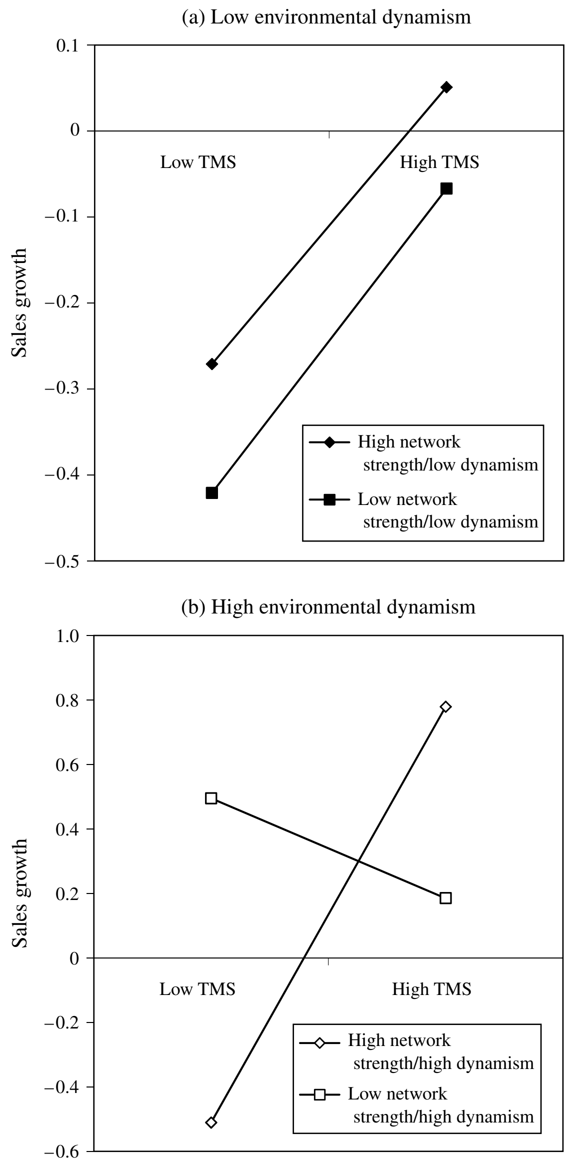
To ensure the fidelity of the findings and the veracity of our implications, we took several methodological and statistical steps to eliminate the possible influence of common method bias. First, to mitigate any potential bias caused by common method effects, we employed several procedural preemptive techniques: using separate respondents to measure the independent and dependent variables, using different data collection modes to gather CEO and TMT respondents, ensuring anonymity by building rapport with respondents, carefully ordering questions to avoid causal interpretations, and adapting scale items to the research context (Podsakoff et al. 2003). The independent and dependent variables were measured using separate respondents. TMT responses were used to measure the independent construct, transactive memory, whereas CEO responses were used to

Figure 3 **Three-Way Interaction of TMT Transactive Memory, TMT External Tie Strength, and Environmental Dynamism for Performance Rating**



measure our dependent constructs, performance rating and sales growth. In most instances, these responses were collected two or three weeks apart using different modes of survey execution. The CEO responses were collected by means of an on-site personal interview, where the first author was able to reassure the CEO of the confidentiality of the data provided. This was essential to gaining accurate measures of sales growth. The TMT responses were completed independently and mailed back.[3] The use of multiple respondents is particularly helpful in minimizing the potential of social desirability bias affecting the relationship between the independent and dependent variables; it also rules out the influence of consistency motives (Podsakoff et al. 2003). We also took other steps to lessen the potential for common method bias. These included ensuring

Figure 4 **Three-Way Interaction of TMT Transactive Memory, TMT External Tie Strength, and Environmental Dynamism for Sales Growth**



respondent anonymity to limit social desirability incentives and counterbalancing questions to reduce consistency motives.

Even as the above procedures effectively preempt common method concerns, we performed several additional tests to detect any residual sources of common method variance. First, we performed Harman's single factor test. An unrotated principal components factor analysis revealed eight factors with eigenvalues greater than 1.0, which together account for 69% of the total variance. Second, to account for any potential influence of common method bias, we employed partial correlation procedures in which a measure of the assumed source of method variance is included as a covariate in the statistical analysis. We used the general factor covariate technique by performing an exploratory factor analysis of the main variables in the study. The average factor score for the first three unrotated factors was computed as a control in a supplementary regression model. Because the addition of the factor control did not change the pattern or magnitude of the regression coefficients for the hypothesized effects, our findings are unlikely to be an artifact of common method bias.

Finally, beyond the procedural and statistical safeguards employed, the significance of our two-way and three-way interaction terms indicates that common method bias is unlikely. As (Cummins 1972, p. 657) explains, "There is no a priori reason to suggest that the numerical difference of two variables measured in the same way at two levels of a third variable is influenced by method specificity." Because it has been argued and empirically demonstrated that common method bias dilutes the magnitude of interactions (Evans 1985, Siemsen et al. 2010), the presence of significant interactions suggests that our results are not likely biased by common method effects.

5.4. Alternative Model Specifications

We further tested the sturdiness of our findings in two ways. First, because the CEO yields a disproportionate influence on the inner dynamics and functioning of the TMT, our exclusion of CEOs' evaluations of TMTs' transactive memory might have biased our measurement and findings. To test this possibility, we created a measure of transactive memory by combining both CEO and TMT assessments. Using this aggregated measure, we replicated the regression analysis reported earlier. The results were unchanged, suggesting that the finding is not an artifact of our measurement approach. Second, although we obtained a relatively high intrateam response rate (close to 60%) and observed high levels of interrater agreement and interrater reliability for all aggregated measures, we also examined whether the findings are robust to different levels of TMT member participation in the survey. In particular, we reanalyzed all models for a subsample of firms in which we excluded those cases where we received only two responses (i.e., the CEO and TMT members). Our results for this subsample were weaker but of the same pattern as the full sample.

6. Discussion

Transactive memory has emerged as an exciting and versatile concept with significant promise for explaining the behavior, functioning, and performance of teams. Although much has already been learned about the development and functioning of transactive memory in work teams, limited progress has been made in extending the concept into the domain of upper echelons theory and the study of TMTs. Yet transactive memory not only offers a new theoretical lens for the study of cognition in TMTs but also affords a unique opportunity to develop a novel understanding of the contingencies and constraints that shape transactive memory effects in this setting. We began with the insight that, owing to the multifaceted locus of top managers' work at the firm-environment boundary, the influence of a TMT's transactive memory on firm performance is likely complicated by contextual contingencies that span the boundary. In particular, we shed new light on how the influence of a TMT's transactive memory is shaped by both the strength of top managers' ties to external actors and the rate of dynamism in the environment. We found support for this argument using a multiple-respondent, multiple-outcome-based survey design of 99 SMEs in technology-intensive industries. In the next section, we outline the implications of our study for relevant theory and research.

6.1. Theoretical Implications

6.1.1. Implications for Upper Echelons Theory. For upper echelons theory, the baseline finding of a positive association between TMT transactive memory and firm performance is an important extension. Above and beyond the compositional (e.g., average tenure, average tenure functional diversity) and structural (i.e., size) context of the team, the development of TMT transactive memory by which teams organize, focus, and expand their cognitive resources to address task demands has a significant association with firm performance. This finding, which was observed across two alternative measures of firm performance while controlling for past performance, suggests that the concept of transactive memory can enrich upper echelons theorizing and testing. In particular, a focus on transactive memory helps to address one of the cognitive "black boxes" of research on top managers—the question of how they respond to strategic situations that exceed the collective abilities of individual team members. Even as scholars have long maintained that the logic of upper echelons theory only holds to the extent that a TMT engages in collective information processing, the question of how these teams organize, expand, and harness their cognitive resources has remained elusive (Hambrick 2007). And although the behavioral integration concept has provided conceptual leverage, it was never intended to represent the cognitive structure of TMTs, nor does it explain how teams distribute and expand their cognitive resources without intensive communication. On the basis of the initial evidence presented here, we thus conclude that transactive memory can impart a powerful theoretical and empirical approach to advance upper echelons research.

A second implication for upper echelons theory is that the external network of top managers does not operate independently of the TMT's cognitive structure. Our two-way and three-way interactional findings reveal that managerial cognitions and social networks play complementary roles; whereas cognition functions to organize the team's knowledge resources, external ties serve to keep these knowledge processes aligned with external events. Beyond this complementarity, our findings also provide cognitive insights into the question of whether and under what conditions maintaining strong ties is more or less beneficial for top managers and their organizations. Although our findings suggest that maintaining strong external ties in the presence of a well-developed transactive memory is advantageous, a closer inspection of Figures 1 and 2 indicates that without a well-developed transactive memory, performance is higher among those TMTs that have avoided forming strong ties with external actors and institutions. An inspection of panel (b) in Figures 3 and 4 indicates that this pattern of effects is pronounced at high levels of environmental dynamism, where TMTs with low transactive memory perform better without strong external ties.

Without additional data on the underlying generative processes, it is difficult to make any conclusive inferences about this pattern of findings. One possible explanation is that the ability of top managers to absorb the knowledge benefits and suppress the costs of strong ties in dynamic settings depends on having a system of cognitive coordination among top managers. Transactive memory may provide a cognitive structure for renewing, updating, and transforming the knowledge of a TMT rapidly, which dynamic settings often require. Thus, strong ties are more likely to be beneficial in dynamic as opposed to stable settings. However, in the absence of a sufficiently developed system of transactive memory, the information-processing costs of strong ties may exceed the benefits. This is because the lack of an efficient system for encoding, storing, and retrieving knowledge impedes the ability of managers to organize, absorb, and harness the full knowledge benefits of strong external ties. Particularly given the rate and unpredictability of environmental change, the lack of a system for processing knowledge flows invariably increases the informational and cognitive burden on each manager, who must struggle to channel these flows from external actors across multiple knowledge domains. Another explanation is that a focus on strong ties in the absence of a well-developed TMS may hinder the ability of the top management team to align internal and external interdependencies. Although managing the internal and external domains of the organization is a central managerial responsibility, internal and external activities compete for top managers' time, attention, and resources (Choi 2002). Without an efficient system for managing the TMT's cognitive demands, managers might be forced into trade-offs between internal and external activities. Thus, in the absence of a TMS, strong external ties may possibly come at the expense of managing internal team relationships as well as the broader task of managing multifunctional dynamics across the organization.[4] We would hasten to add that further research is needed to more fully tease out these effects.

6.1.2. Transactive Memory Research. For theory and research on transactive memory, our findings offer renewed theoretical insights into the contingent nature of the relationship between transactive memory and performance. In particular, they cast light on some of the conditions under which a TMT's transactive memory is effectual, ineffectual, and counterproductive. Although there has been some research into the moderators of transactive memory effects in other team settings, the focus has mostly been on internal team- and task-centered contingencies. The influence of the external environment and teams' connectedness to that environment has not received systematic attention from transactive memory researchers. Yet our examination of transactive memory in the TMT domain suggests that such contingencies are theoretically and empirically significant. First, our findings suggest that the value of a TMT's transactive memory is tied to how strongly members are connected to actors and institutions outside the firm. Furthermore, our analyses reveal that not all networks are created equally, and the strength of the connections appears to be a determining factor in whether they enhance the performance effects of a transactive memory. Without a strong set of ties to customers, suppliers, and other actors and institutions, the contribution of a transactive memory to a firm's performance is significantly diminished. For theory, this suggests that the inner cognitive workings of transactive memory are complemented by boundary-spanning activities, particularly the cultivation of strong ties outside the TMT. The development of such a support system serves as the basis for updating the expertise of team members, because strong ties are essential to the transfer of knowledge. It also serves to maintain the relevance and accuracy of knowledge within the team. This is an important finding for transactive memory research, which has been relatively silent on the external mechanisms that can enable team members to update their knowledge and how such mechanisms may shape the ultimate influence of transactive memory.

A deeper examination of this effect further shows that the facilitative influence and potency of strong ties grow proportionally with increased flux in market and technological environment(s). A test of the three-way interaction among transactive memory, the average strength of external ties, and environmental dynamism indicated that strong ties play a more prominent role when the environment is dynamic rather than stable. As shown in Figures 3 and 4, the interaction between transactive memory and strong ties becomes most pronounced at high levels of environmental dynamism. At low levels of dynamism, the effect of transactive memory on performance does not appear to depend on the strength of ties. This may be because the knowledge repertoire of members does not need to be updated as frequently. Another possible explanation is that the TMT's existing frames of reference remain valid in stable settings. This is an important finding, partly because of the costs and constraints associated with maintaining strong external ties (Adler and Kwon 2002) but also because it appears that under stable conditions, a loose coupling with external actors and institutions may be sufficient.

However, in a fluctuating environment the average strength of external ties becomes instrumental to the performance potential of transactive memory. In particular, panel (b) in Figures 3 and 4 reveals a number of important insights about the relationship between transactive memory, external network strength, and firm performance under conditions of high environmental dynamism. First, when network strength is low, the relationship between TMS and both objective and subjective performance becomes negative. The interaction plots appear to indicate that teams with low TMS and low network strength perform better than teams with high TMS and low network strength. This raises the question, why does TMS become counterproductive when external network strength is low? One explanation is that in dynamic settings a TMS without conduits of access to timely and preferential insights becomes a barrier to developing new and novel interpretations of the environment. Without strong external connections, TMS may give rise to information exchange that is less timely, relevant, and distinctive and that may be out of sync with external events and developments. Thus, teams with a high rather than low TMS become disadvantaged in such situations because the routinized and embedded nature of informational exchanges becomes a source of cognitive inertia. Members continue to rely on each other to stay current, even as the informational and interpretive base of the team becomes dislodged from the outside environment. Thus, in the absence of strong ties, a TMS may become maladaptive, decoupling the team's interpretational repertoire from the market. For upper echelons researchers, this finding will come as no surprise. Executives spend a great deal of time engaged in boundary-spanning activities, and the value of their work is often driven by external influences. However, this finding is significant for transactive memory researchers in that it addresses a key call in transactive memory research that "establishing the limits of the TMS construct as well as its applicability in different types of organizational teams are key areas for future research" (Lewis 2003, p. 602).

A second, somewhat counterintuitive observation is that under conditions of high dynamism, those teams with high TMS/high network strength perform at a comparable level to those with low TMS/low network strength, particularly with respect to our subjective performance measure. One tentative explanation is that different configurations of internal cognitive systems and external networks maybe appropriate under different types of dynamic conditions. Given the wide technological and market scope of firms in our sample, the source, nature, and implications of dynamism are likely to vary considerably across firms, which may have implications for the optimal alignment of internal cognitive systems and external networks. The combination of low TMS and low network strength, for example, might be beneficial to those operating in dynamic contexts where a dominant design or technical standard has yet to emerge, where the fast-shifting landscape calls for flexibility, both cognitively in terms of the specialized knowledge and competence needed within the team and relationally in terms of network needs. Establishing a TMS and strong external ties in such settings may be counterproductive given fundamental uncertainties about the trajectory of technological development and the locus of relevant knowledge. Although this finding may point to a certain degree of equifinality in the relationship between transactive memory, external network strength, and firm performance, these are obviously tentative insights, and as such, further research employing more intrusive methods may be needed to unravel the complex configuration of transactive memory and network structure most beneficial to firm performance across environmental conditions.

6.2. Future Research Implications

The findings suggest some promising avenues for future research. Given our main effect finding, an immediate priority for future research is to examine the generative mechanisms that underlie the relationship between a TMT's transactive memory and firm performance. Although we predicted that transactive memory would improve environmental scanning, strategic issue diagnosis, and decision making, these expectations require empirical confirmation—preferably in a longitudinal field setting. Second, it could be particularly informative to build insight into the development and emergence of transactive memory in TMTs, especially using qualitative studies of a longitudinal nature. Although prior research has discussed the development of transactive memory in work teams, TMTs might display additional patterns of development and emergence, as executive contexts can place significant demands on the development and maintenance of cognitive interdependency even if cognitive resources are adequate within the team. Top teams can display a range of dysfunctions—such as frequent turnovers, succession tournaments, and functional turf wars—that could thwart the development and functioning of transactive memory. Because members are typically responsible for distinct organizational functions and units, and they compete for scarce resources and attention, some top teams may even develop dysfunctional transactive memory—a sort of transactive “baggage” memory—from these relational conflicts. The defeat of a favored strategic initiative could leave dark spots in some members' transactive memory for years to come. And although transactive memory research on lower-level teams provides some initial insights for these examinations, we suspect that Hambrick's (1995) discussion on the antecedents of sociopolitical fragmentation within top teams can provide a meaningful foundation to begin exploring productive (e.g., cognitive division of labor), unproductive (e.g., excessive reliance on others), and destructive (e.g., baggage memory) transactive memory in TMTs.

We acknowledge, too, that our focus here is on only one aspect of a complex set of conditioning factors on the relationship between transactive memory and firm performance. Given the externally oriented nature of executive work, we sought and found evidence suggesting that a top team's external network of ties can shape the effect of transactive memory on firm performance. Within the TMT literature, the notion and determinants of managerial discretion could particularly point to additional contingencies. Finally, SME firms differ from their larger counterparts in that the former have fewer intervening levels of management and are less constrained by extraneous influences. Thus, although our findings offer a reasonable point of departure from large public firms, future research is needed to examine their veracity in that context. Such efforts would constitute what Eden (2002) called a constructive replication that utilizes different methods and samples to strengthen the confidence in the validity of an original finding.

6.3. Limitations and Conclusion

Even as our research design ruled out common method bias, an experimental/longitudinal design would naturally be preferred to advance stronger causal claims. However, we are encouraged by the “objective” nature of our moderating variable as well as by consistent results for both subjective and objective measures of performance. To mitigate potential measurement errors, we used pretested and valid constructs and performed extensive testing to demonstrate discriminant validity. Furthermore, we are reasonably confident in the veracity of our findings given that we replicated them with a quasi-objective measure of sales growth. We would be remiss if we did not also mention the existence of several distinct strengths in using multifaceted performance measures as assessed by executives relative to direct competitors. Indeed, although our results were consistent across both measures of performance, the findings were generally stronger for performance rating than for sales growth. In particular, the moderating effect of strength of external ties and environmental dynamism, as well as the main effect of environmental dynamism, was stronger for our subjective rating of performance than our sales growth measure. Although the differential magnitude of effects is likely an artifact of the broader multidimensional measure of subjective performance, it may also reflect differences in the potency of effects for relative versus absolute performance. Because our subjective measure captures differences in performance relative to that of competitors, it is likely more sensitive to the interactive influence of transactive memory and strength of external ties. This is because the private and preferential knowledge bestowed by strong ties helps to further differentiate the team's knowledge processes and inputs from those of competitors. We also note the likelihood that dimensions of performance can be differently shaped by a TMT's transactive memory. Therefore, it is important for researchers to be cognizant of this variability in interpreting our results, particularly when counseling actual managers and organizations (Miller et al. 2013). Although our study provides a fulcrum on which such insights might be leveraged, we also have barely scratched the surface in terms of the range of such social network and contextual variables that might shape the transactive memory–performance relationship.

In closing, our study enriches both the transactive memory and upper echelons literatures by demonstrating that the effect of transactive memory in the TMT setting cannot be separated from the external context in which these teams are embedded, in ways that complicate the operation, magnitude, and direction of its influence. Although the effects of a TMT's transactive memory on performance seem largely positive, our findings help to deconstruct when transactive memory is likely beneficial and cast new insight into when it can be potentially detrimental. We hope these insights and findings pave the way for richer explanations of both the promise and perils of extending the concept of transactive memory into the domain of upper-echelon executives.

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Endnotes

[1]CorpTech provides data on sales growth for some companies. However, the data are subject to three limitations. First, the data have poor coverage—less than 10% of the firms in our sample had data on sales growth in the CorpTech database. Second, although these data were collected by telephone interview, it is not clear whether the sales data come from a credible or knowledgeable source. Third, for most companies in the CorpTech database, the sales data are not updated annually. All this is to say that we place more faith in our sales growth data, which were collected on site during an interview with the CEO.

[2]We also examined partial scatterplots for each variable in our regression analysis to ensure that the pattern of the partial effect for each interaction matched the pattern depicted in our interaction plots.

[3]In some cases, we arranged to collect the survey packets on site.

[4]We thank an anonymous reviewer for this insight.

References

Adler PS, Kwon SW (2002) Social capital: Prospects for a new concept. *Acad. Management Rev.* 27(1):17–40.

Aguilar FJ (1967) *Scanning the Business Environment* (Macmillan, New York).

Ahuja G, Lampert CM (2001) Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management J.* 22(6–7):521–543.

Aiken LS, West SG (1991) *Multiple Regression: Testing and Interpreting Interactions* (Sage, Newbury Park, CA).

Akgun AE, Byrne JC, Keskin H, Lynn GS (2006) Transactive memory system in new product development teams. *IEEE Trans. Engrg. Management* 53(1):95–111.

Amason AC, Sapienza HJ (1997) The effects of top management team size and interaction norms on cognitive and affective conflict. *J. Management* 23(4):495–516.

Ancona DG, Caldwell DF (1992) Demography and design: Predictors of new product team performance. *Organ. Sci.* 3(3):321–341.

Anderson JC, Gerbing DW (1988) Structural equation modeling in practice—A review and recommended 2-step approach. *Psych. Bull.* 103(3):411–423.

Anderson P, Tushman ML (2001) Organizational environments and industry exit: The effects of uncertainty, munificence and complexity. *Indust. Corporate Change* 10(3):675–711.

Atuahene-Gima K, Li HY (2004) Strategic decision comprehensiveness and new product development outcomes in new technology ventures. *Acad. Management J.* 47(4):583–597.

Austin JR (2003) Transactive memory in organizational groups: The effects of content, consensus, specialization, and accuracy on group performance. *J. Appl. Psych.* 88(5):866–878.

Bagozzi RP, Yi YJ, Phillips LW (1991) Assessing construct-validity in organizational research. *Admin. Sci. Quart.* 36(3):421–458.

Bliese PD (2000) Within-group agreement, non-independence, and reliability: Implications for data aggregation and analysis. Klein KJ, Kozlowski SWJ, eds. *Multilevel Theory, Research, and Methods in Organizations: Foundations, Extensions, and New Directions* (Jossey-Bass, San Francisco), 349–381.

Bunderson JS, Sutcliffe KM (2002) Comparing alternative conceptualizations of functional diversity in management teams: Process and performance effects. *Acad. Management J.* 45(5):875–893.

Buyl T, Boone C, Matthyssens P (2011) Upper echelons research and managerial cognition. *Strategic Organ.* 9(3):240–246.

Cannella AA Jr, Park J-H, Lee H-U (2008) Top management team functional background diversity and firm performance: Examining the roles of team member colocation and environmental uncertainty. *Acad. Management J.* 51(4):768–784.

Cao Q, Simsek Z, Zhang HP (2010) Modelling the joint impact of the CEO and the TMT on organizational ambidexterity. *J. Management Stud.* 47(7):1272–1296.

Chan D (1998) Functional relations among constructs in the same content domain at different levels of analysis: A typology of composition models. *J. Appl. Psych.* 83(2):234–246.

Choi JN (2002) External activities and team effectiveness review and theoretical development. *Small Group Res.* 33(2):181–208.

Cohen J, Cohen P, West SG, Aiken LS (2003) *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences* (Lawrence Erlbaum Associates, Mahwah, NJ).

Collins CJ, Clark KD (2003) Strategic human resource practices, top management team social networks, and firm performance: The role of human resource practices in creating organizational competitive advantage. *Acad. Management J.* 46(6):740–751.

Cummins RC (1972) Leader-member relations as a moderator of the effects of leader behavior and attitude. *Personnel Psych.* 25(4):655–660.

Daft RL, Weick KE (1984) Toward a model of organizations as interpretation systems. *Acad. Management Rev.* 9(2):284–295.

Danneels E, Sethi R (2011) New product exploration under environmental turbulence. *Organ. Sci.* 22(4):1026–1039.

Dawson JF, Richter AW (2006) Probing three-way interactions in moderated multiple regression: Development and application of a slope difference test. *J. Appl. Psych.* 91(4):917–926.

Dess GG, Robinson RB (1984) Measuring organizational performance in the absence of objective measures—The case of the privately-held firm and conglomerate business unit. *Strategic Management J.* 5(3):265–273.

Eden D (2002) Replication, meta-analysis, scientific progress, and *AMJ*'s publication policy. *Acad. Management J.* 45(5):841–846.

Eisenhardt KM (1989) Making fast strategic decisions in high-velocity environments. *Acad. Management J.* 32(3):543–576.

Evans MG (1985) A Monte Carlo study of the effects of correlated method variance in moderated multiple regression analysis. *Organ. Behav. Human Decision Processes* 36(3):305–323.

Faraj S, Sproull L (2000) Coordinating expertise in software development teams. *Management Sci.* 46(12):1554–1568.

Faraj S, Xiao Y (2006) Coordination in fast-response organizations. *Management Sci.* 52(8):1155–1169.

Forbes DP (2007) Reconsidering the strategic implications of decision comprehensiveness. *Acad. Management Rev.* 32(2):361–376.

Fornell C, Larcker DF (1981) Evaluating structural equation models with unobservable variables and measurement error. *J. Marketing Res.* 18(1):39–50.

Geletkanycz MA, Hambrick DC (1997) The external ties of top executives: Implications for strategic choice and performance. *Admin. Sci. Quart.* 42(4):654–681.

Granovetter M (1973) Strength of weak ties. *Amer. J. Sociol.* 78(6):1360–1380.

Hambrick DC (1989) Putting top managers back in the strategy picture. *Strategic Management J.* 10(S1):5–15.

Hambrick DC (1994) Top management groups: A conceptual integration and reconsideration of the "team" label. Staw BM, Cummings LL, eds. *Research in Organizational Behavior*, Vol. 16 (JAI Press, Greenwich, CT), 171–213.

Hambrick DC (1995) Fragmentation and the other problems CEOs have with their top management teams. *Calif. Management Rev.* 37(3):110–127.

Hambrick DC (2007) Upper echelons theory: An update. *Acad. Management Rev.* 32(2):334–343.

Hambrick DC, Fukutomi GDS (1991) The seasons of a CEOs tenure. *Acad. Management Rev.* 16(4):719–742.

Hambrick DC, Mason PA (1984) Upper echelons—The organization as a reflection of its top managers. *Acad. Management Rev.* 9(2):193–206.

Hansen MT (1999) The search-transfer problem: The role of weak ties in sharing knowledge across organization subunits. *Admin. Sci. Quart.* 44(1):82–111.

He Z-L, Wong P-K (2004) Exploration vs. exploitation: An empirical test of the ambidexterity hypothesis. *Organ. Sci.* 15(4):481–494.

Heavey C, Simsek Z (2014) Distributed cognition in top management teams and organizational ambidexterity: The influence of transactive memory systems. *J. Management*, ePub ahead of print August 20, http://dx.doi.org/10.1177/0149206314545652.

Heavey C, Simsek Z, Fox BC (2015) Managerial social networks and ambidexterity of SMEs: The moderating role of a proactive commitment to innovation. *Human Resource Management*, ePub ahead of print March 2, http://dx.doi.org/10.1002/hrm.21703.

Henderson AD, Miller D, Hambrick DC (2006) How quickly doe CEOs become obsolete? Industry dynamism, CEO tenure, and company performance. *Strategic Management J.* 27(5): 447–460.

Hollingshead AB (2000) Perceptions of expertise in and transactive memory in work relationships. *Group Processes Intergroup Relations* 3(3):257–267.

Huber GP, Daft RL (1987) The information environments of organizations. Jablin FM, Putnam LL, Roberts KH, Porter LW, eds. *Handbook of Organizational Communication: An Interdisciplinary Perspective* (Sage, Thousand Oaks, CA), 130–164.

Ingram P, Roberts PW (2000) Friendships among competitors in the Sydney hotel industry. *Amer. J. Sociol.* 106(2):387–423.

James LR, Demaree RG, Wolf G (1984) Estimating within-group interrater reliability with and without response bias. *J. Appl. Psych.* 69(1):85–98.

James LR, Demaree RG, Wolf G (1993) R_{wg}: An assessment of within-group interrater agreement. *J. Appl. Psych.* 78(2): 306–309.

Janis IL (1982) *Groupthink: Psychological Studies of Policy Decisions and Fiascoes* (Houghton Mifflin, Boston).

Kilduff M, Angelmar R, Mehra A (2000) Top management-team diversity and firm performance: Examining the role of cognitions. *Organ. Sci.* 11(1):21–34.

Kline RB (1998) *Principles and Practice of Structural Equation Modeling* (Guilford Press, New York).

LeBreton JM, Senter JL (2008) Answers to 20 questions about interrater reliability and interrater agreement. *Organ. Res. Methods* 11(4):815–852.

Lewis K (2003) Measuring transactive memory systems in the field: Scale development and validation. *J. Appl. Psych.* 88(4): 587–604.

Lewis K (2004) Knowledge and performance in knowledge-worker teams: A longitudinal study of transactive memory systems. *Management Sci.* 50(11):1519–1533.

Lewis K, Herndon B (2011) Transactive memory systems: Current issues and future research directions. *Organ. Sci.* 22(5): 1254–1265.

Lewis K, Lange D, Gillis L (2005) Transactive memory systems, learning, and learning transfer. *Organ. Sci.* 16(6):581–598.

Liang DW, Moreland R, Argote L (1995) Group versus individual training and group-performance—The mediating role of transactive memory. *Personality Soc. Psych. Bull.* 21(4):384–393.

Littlepage GE, Hollingshead AB, Drake LR, Littlepage AM (2008) Transactive memory and performance in work groups: Specificity, communication, ability differences, and work allocation. *Group Dynamics: Theory, Res., Practice* 12(3):223–241.

Lubatkin MH, Simsek Z, Ling Y, Veiga JF (2006) Ambidexterity and performance in small- to medium-sized firms: The pivotal role of top management team behavioral integration. *J. Management* 32(5):646–672.

McDonald ML, Westphal JD (2003) Getting by with the advice of their friends: CEOs' advice networks and firms' strategic responses to poor performance. *Admin. Sci. Quart.* 48(1):1–32.

McDonald ML, Khanna P, Westphal JD (2008) Getting them to think outside the circle: Corporate governance, CEOs' external advice networks, and firm performance. *Acad. Management J.* 51(3):453–475.

Mehra A, Dixon AL, Brass DJ, Robertson B (2006) The social network ties of group leaders: Implications for group performance and leader reputation. *Organ. Sci.* 17(1):64–79.

Milkovich GT (1987) Compensation systems in high technology companies. Balkin DB, Gomez-Mejia LR, eds. *New Perspectives on Compensation* (Prentice-Hall, Englewood Cliffs, NJ), 269–277.

Miller CC (2008) Decisional comprehensiveness and firm performance: Towards a more complete understanding. *J. Behav. Decision Making* 21(5):598–620.

Miller CC, Washburn NT, Glick WH (2013) The myth of firm performance. *Organ. Sci.* 24(3):948–964.

Narayanan VK, Zane LJ, Kemmerer B (2011) The cognitive perspective in strategy: An integrative review. *J. Management* 37(1): 305–351.

Neter J, Kutner MH, Nachtsheim CJ, Wasserman W (1996) *Applied Linear Statistical Models*, 4th ed. (Irwin, Chicago).

Organ DW (1971) Linking pins between organizations and environment. *Bus. Horizons* 14(6):73–80.

Peltokorpi V (2008) Transactive memory systems. *Rev. General Psych.* 12(4):378–394.

Podsakoff PM, MacKenzie SB, Lee JY, Podsakoff NP (2003) Common method biases in behavioral research: A critical review of the literature and recommended remedies. *J. Appl. Psych.* 88(5):879–903.

Rau D (2005) The influence of relationship conflict and trust on the transactive memory—Performance relation in top management teams. *Small Group Res.* 36(6):746–771.

Rau D (2006) Top management team transactive memory, information gathering, and perceptual accuracy. *J. Bus. Res.* 59(4): 416–424.

Ren Y, Argote L (2011) Transactive memory systems 1985–2010: An integrative framework of key dimensions, antecedents, and consequences. *Acad. Management Ann.* 5(1):189–229.

Ren Y, Carley KM, Argote L (2006) The contingent effects of transactive memory: When is it more beneficial to know what others know? *Management Sci.* 52(5):671–682.

Richard PJ, Devinney TM, Yip GS, Johnson G (2009) Measuring organizational performance: Towards methodological best practice. *J. Management* 35(3):718–804.

Robinson KC, McDougall PP (2001) Entry barriers and new venture performance: A comparison of universal and contingency approaches. *Strategic Management J.* 22(6–7):659–685.

Robinson RB, Pearce JA (1988) Planned patterns of strategic behavior and their relationship to business-unit performance. *Strategic Management J.* 9(1):43–60.

Shimizu K, Hitt MA (2004) Strategic flexibility: Organizational preparedness to reverse ineffective strategic decisions. *Acad. Management Executive* 18(4):44–59.

Siemsen E, Roth A, Oliveira P (2010) Common method bias in regression models with linear, quadratic, and interaction effects. *Organ. Res. Methods* 13(3):456–476.

Smith KG, Collins CJ, Clark KD (2005) Existing knowledge, knowledge creation capability, and the rate of new product introduction in high-technology firms. *Acad. Management J.* 48(2):346–357.

Stam W, Elfring T (2008) Entrepreneurial orientation and new venture performance: The moderating role of intra- and extraindustry social capital. *Acad. Management J.* 51(1):97–111.

Tabachnick BG, Fidell LS (1996) *Using Multivariate Statistics*, 3rd ed. (HarperCollins, New York).

Tushman M, Romanelli E (1985) Organizational evolution: A metamorphosis model of convergence and reorientation. Cummings LL, Staw BM, eds. *Research in Organizational Behavior*, Vol. 7 (JAI Press, Greenwich, CT), 171–222.

Uzzi B, Lancaster R (2003) Relational embeddedness and learning: The case of bank loan managers and their clients. *Management Sci.* 49(4):383–399.

Wegner DM (1987) Transactive memory: A contemporary analysis of the group mind. Mullen B, Goethals GR, eds. *Theories of Group Behavior* (Springer, New York), 185–208.

Wegner DM, Giuliano T, Hertel P (1985) Cognitive interdependence in close relationships. Ickes WJ, ed. *Compatible and Incompatible Relationships* (Springer, New York), 253–276.

Weick KE, Roberts KH (1993) Collective mind in organizations: Heedful interrelating on flight decks. *Admin. Sci. Quart.* 38(3):357–381.

Williams JR (1992) How sustainable is your competitive advantage? *Calif. Management Rev.* 34(3):29–51.

Zheng YF (2012) Unlocking founding team prior shared experience: A transactive memory system perspective. *J. Bus. Venturing* 27(5):577–591.

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